

Wilson, Chow Sin Long

sinlongchow@gmail.com · slchow.com · linkedin.com/in/wilsonslchow · github.com/ocwilsonchow

Full-Stack Software Engineer · AWS Certified Solutions Architect – Associate

SUMMARY

Full-stack **software engineer** with **5 years of experience** delivering **15+** React and React Native products from Figma through Node.js APIs and AWS, including Apple App Store and Google Play releases. Cut load times by **80%** and search latency by **90%**. Now leading a multi-channel **Generative AI** platform with durable workflows, memory, RAG, and evals. **AWS Certified Solutions Architect – Associate**.

EXPERIENCE

Software Developer, Vioo Company Limited

May 2022 – Present

Primary engineer for concurrent client products, owning delivery from UX scoping through cloud deployment.

- Shipped and maintained **15+** React / React Native apps with high-fidelity Figma design, shared design systems, and Apple App Store / Google Play releases.
- Migrated WordPress sites to Strapi / Sanity and Meiliseach / Algolia in a **Turborepo** on **SST** without interrupting publishing; cut load times by **80%**, search latency by **90%**, and achieved PageSpeed scores above **95**.
- Built typed Node.js APIs with Hono, OpenAPI, and tRPC on Lambda / API Gateway and **ECS Fargate**; modeled org-scoped Postgres with Drizzle / Prisma, Better Auth, OAuth 2.0 / OIDC, and RBAC.
- Owned shared **Radix / Tailwind / shadcn/ui** component and hook APIs; **Storybook**, Fumadocs, and MDX guidance eliminated duplicate UI across client apps.
- Set **Vitest**, React Testing Library, and **Playwright / Cypress** quality gates for shared UI and critical flows.
- Instrumented web, mobile, and API releases with **Sentry** for production crashes, device issues, and API errors.
- Adopted **tRPC** and Scalar-generated API docs, cutting integration time by **50%** and eliminating schema-mismatch errors.
- Built GitHub Actions + **SST** CI/CD with linting, type-checking, tests, and builds; cut deployment time by **50%+**.
- Mentored interns and junior developers through pairing, code reviews, and production delivery.

SELECTED PROJECTS

Multi-Channel Agentic Platform

Apr 2026 – Present

Durable agent that empowers internal tools and workflows.

- Built agent runs durable streams so clients could disconnect, reconnect, and resume without losing progress.
- Built **Mastra workflows**, memory, schedules, tools, pause/resume, retries, sandboxing, and skills.
- **RAG** (ingestion, chunking, **Pinecone** query/reranking); **SQL agent** to query database in natural language.
- Ran Hono + Better Auth on **ECS Fargate**, with Postgres, S3, and Mastra Platform for observability.
- Built a custom **@mentions-style** input component for attaching workspace files to messages.

Healthcare AI Agent

July 2025 – Apr 2026

Healthcare and shopping assistant grounded in uploaded documents and Shopify store product data.

- Questionnaires/health report -> structured insights -> personalized product recommendations -> add to product cart.
- Owned document upload, questionnaires, product/cart actions, and admin flows.
- Built backend with **Hono, LangChain, Qdrant, Better Auth**; Frontend with **Next.js, AI SDK**.
- Built rich chatbot tool invocation UIs such as in-thread file upload, product cards, cart actions, multi-step questionnaires, and data charts.
- Enforced application security with **HttpOnly cookies, CORS**, and **RBAC**.
- Deployed on AWS with IaC for scalable, production-ready infrastructure.

High-Performance Editorial Platform

Nov 2025

Strapi, Meiliseach, and Next.js editorial platform in the WordPress-to-headless migration.

- Reserved image space and optimized caching/images: **LCP < 0.5s, CLS < 0.1, PageSpeed >90** (lab).
- Used hybrid ISR: main pages/latest 100 articles at build, rest on demand; Strapi **GraphQL, Next.js** cache tags, and webhooks revalidated publishes without full rebuilds.
- Made keyboard navigation, accessible form errors, semantics, and Lighthouse/axe checks release requirements.
- Split interactive state from page data: **XState + TanStack Form / Zod**, route-owned locale, and hybrid-ISR content.
- Built a Strapi plugin for AI content generation with AI SDK.

Restaurant Membership Platform

shipped Apr 2024

React Native membership app, paired with a Express backend and Next.js admin portal.

- Owned application delivery from Figma, **React Native** development, through Apple App Store / Google Play release; maintained backend API and admin portal.
- Maintained **99%** crash-free sessions by diagnosing crashes, device issues, and API bottlenecks with **Sentry**.
- Fixed JS-thread feed stalls with **FlashList**, memoized rows, smaller cached images, and native-thread **Reanimated** / **Moti**; froze/deferred inactive tabs, ensured consistent **60 FPS** performance.
- Routed Firebase Cloud Messaging promo/milestone deep links to the matching screen.

OTHER WORK

Also shipped React Native student and restaurant loyalty apps, a Flutter elderly-care app, Shopify storefronts, CMS sites, a Skia photobooth, and multiple RBAC admin systems.

SKILLS

Languages: TypeScript, Node.js, SQL, Python

Design: Figma

Frontend: React, Next.js, React Native, Expo, HTML/CSS, Tailwind CSS, shadcn/ui, TipTap

UI behavior: TanStack Query, TanStack Form, React Hook Form, XState, TanStack Table, next-intl

Architecture: state modeling (URL/server/local), component APIs, routing, SSR/ISR, GraphQL, tRPC, OpenAPI clients, Storybook, Fumadocs

Backend: Hono, Drizzle, Prisma, Postgres, DynamoDB, Better Auth

Auth & security: OAuth 2.0/OIDC, JWT, HttpOnly cookies, CORS, RBAC

AI: Mastra, Vercel AI SDK, LangChain, Pinecone, Strands, AWS Bedrock

QA: Vitest, Playwright, Cypress, Lighthouse, axe, Sentry

Mobile: FlashList, Reanimated, Moti, Skia, Unistyles

AWS & DevOps: SST, Terraform, Docker, GitHub Actions, AWS, Turborepo

QUALIFICATIONS

- AWS Certified Solutions Architect – Associate, 2026
- AWS Certified Cloud Practitioner, 2025

EDUCATION

University College London

2016 – 2020

- Master of Pharmacy (MPharm) — First Class Honours
- Transitioned from pharmacy to software development after finding passion in building applications.